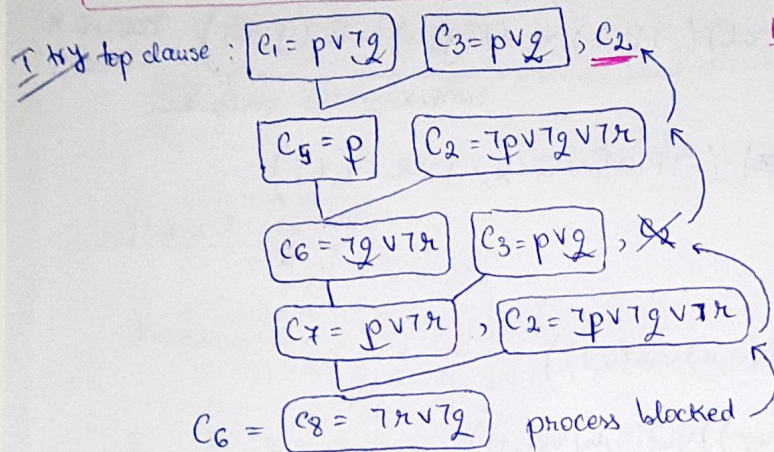


5] Consistency of $S_4 = \{ \underbrace{p \vee \neg q}_{c_1}, \underbrace{\neg p \vee \neg q \vee \neg r}_{c_2}, \underbrace{p \vee q}_{c_3} \}$.

Linear resolution =
= graphical level.

S is inconsistent ($\Rightarrow$) $S \vdash \frac{\text{Lim.}}{\text{Res.}} \square$



Resolution rule:

$$\text{Res}_x(f \vee l, g \vee \neg l) = f \vee g$$

Process blocked when:

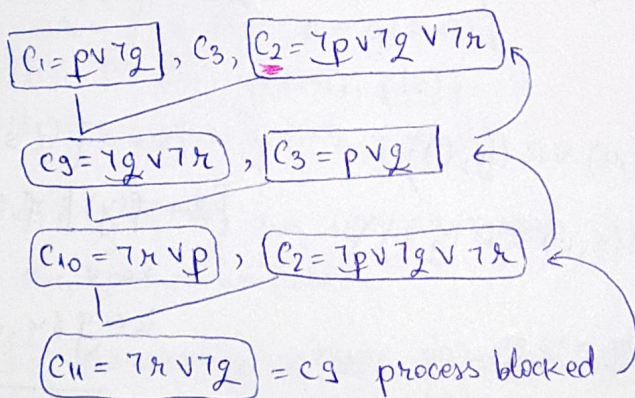
- derive the $\square$
- tautology
- two equal clauses

(all the possible side clauses were used)

→ After a complete search using back strategy, without the derivation of $\square \Rightarrow$ we conclude that S_4 is CONSISTENT.

→ Only after a complete search we can find out consistency.

II try



1.4] Transform the following formulas into prenex, skolem and clausal normal forms.

$$u_4 = (\exists x)(\exists y)((\forall z) \neg P(z) \rightarrow (\forall u) \neg Q(x, u) \rightarrow (\forall z) R(y, z))$$

$$u_4 = (\exists x)(\forall y)((\forall z) P(z) \rightarrow (\forall u) (\neg Q(x, u) \rightarrow (\forall z) R(y, z)))$$

$$u_4 \xrightarrow[\text{1}]{\text{replace}} [(\exists x)(\forall y)((\forall z) \neg P(z) \vee (\forall u) (\neg Q(x, u) \rightarrow (\forall z) R(y, z)))]$$

$$\xrightarrow[\text{2}]{\text{replace}} (\exists x)(\forall y)((\neg (\forall z) P(z)) \vee (\forall u) (Q(x, u) \vee (\forall z) R(y, z)))$$

$$\xrightarrow[\text{De Morgan}]{\text{rename}} (\exists x)(\forall y)((\exists z) \neg P(z) \vee (\forall u) (Q(x, u) \vee (\forall z) R(y, z)))$$

$$\xrightarrow[\text{bound variables}]{\text{rename}} (\exists x)(\forall y)((\exists z) \neg P(z) \vee (\forall u) (Q(x, u) \vee (\forall t) R(y, t)))$$

extract
quantifiers
 (the innermost is t)

We can extract these 2 groups in 2 ways (2!).

$$U_4^{P1} = \underbrace{(\exists x)(\forall y)(\exists z)(\forall u)(\forall t)}_{\text{PREFIX}} \underbrace{(\neg P(z) \vee Q(y,u) \vee R(y,t))}_{\text{MATRIX of the formula.}} \quad \text{PRENEX FORMS.}$$

$$U_4^{P2} = \underbrace{(\exists x)(\forall y)(\forall u)(\forall t)(\exists z)}_{\text{PREFIX}} (\neg P(z) \vee Q(y,u) \vee R(y,t))$$

Skolemisation:

$$U_4^{S1} = (\forall y)(\forall u)(\forall t)(\neg P(a) \vee Q(a,u) \vee R(y,t))$$

$$U_4^{S2} = (\forall y)(\forall u)(\forall t)(\neg P(g(y,u,t)) \vee Q(a,u) \vee R(y,t))$$

Clausal forms:

$$U_4^{C1} = (\neg P(f(y)) \vee Q(a,u) \vee R(y,t))$$

$$U_4^{C2} = (\neg P(f(y,u,t)) \vee Q(a,u) \vee R(y,t))$$

Theoretical results:

a) $U \equiv U^P$

b) U is inconsistent

$(\Rightarrow) U^P \equiv \text{---}$

$(\Rightarrow) U^S \equiv \text{---}$

$(\Rightarrow) U^C \equiv \text{---}$

$[x \leftarrow a]$ a skolem const.

$[z \leftarrow f(y)]$ f skolem funct.

$x \leftarrow a$, a skolem const.

$z \leftarrow g(y,u,t)$

Control, Alt, Reclaim

24. Are the literals from the following pairs unifiable?

$x, y, z \in \text{variables}$

$a, b \in \text{constants}$

$f, g \in F_1$

$P \in P_3$

$\Theta = [x_1 \leftarrow t_1, \dots, x_m \leftarrow t_m]$ substitution.

$$\begin{array}{llll} x \leftarrow a \vee & a \leftarrow x & x \leftarrow f(a) \vee & x \leftarrow f(x) \quad f(x) \neq g \\ x \leftarrow y \vee & a \leftarrow b & x \leftarrow f(y) \vee & f(x) \leftarrow a \end{array}$$

a) $l_1 = P(\underline{a}, g(x), f(g(y)))$ and $l_2 = P(\underline{y}, z, f(z))$
 $\Theta = \varepsilon$

$\text{mgm}(l_1, l_2)?$

it I: $\lambda: [y \leftarrow a]$

(first iteration) $\Theta = \Theta \lambda = [y \leftarrow a]$

$\Theta(l_1) = P(a, g(x), f(g(a)))$

$\Theta(l_2) = P(a, z, f(z))$

it II: $\lambda: [z \leftarrow g(x)]$

(second iteration)

$\Theta = \Theta \lambda = [y \leftarrow a][z \leftarrow g(x)]$

$\Theta(l_1) = P(a, g(x), f(g(a)))$

$\Theta(l_2) = P(a, g(x), f(g(x)))$

itm. $\lambda: [x \leftarrow a]$

$$\theta = \theta\lambda = [y \leftarrow a, z \leftarrow g(x)] [x \leftarrow a] = [y \leftarrow a, z \leftarrow g(x), x \leftarrow a] = \text{mgu}(l_1, l_2)$$

$$\theta(l_1) = \theta(l_2) = P(a, g(a), f(g(a)))$$

COMMON INSTANCE of l_1, l_2 .

MOST GENERAL
UNIFIER of l_1, l_2 .

c) $l_1 = P(x, g(f(a)), f(b))$

$l_2 = P(f(y), z, z)$

$\theta = \varepsilon$

I: $\lambda: [x \leftarrow f(y)]$

$\theta = \theta\lambda = [x \leftarrow f(y)]$

$\theta(l_1) = P(f(y), g(f(a)), f(b))$

$\theta(l_2) = P(f(y), z, z)$

II: $\lambda: [z \leftarrow g(f(a))]$

$\theta = \theta\lambda = [x \leftarrow f(y), z \leftarrow g(f(a))]$

$\theta(l_1) = P(f(y), g(f(a)), f(b))$

$\theta(l_2) = P(f(y), g(f(a)), g(f(a)))$

III: $f(b)$ & $g(f(a))$ are NOT UNIFIABLE (none of them is a variable)
 $\Rightarrow l_1, l_2$ not unifiable.

3.4. Prove the inconsistency of S_4 using LOCK RESOLUTION. (predicate)

$$S_4 = \left\{ \underbrace{P(x) \vee Q(x)}_{(1) \quad (2) \quad C_1}, \underbrace{\neg P(x) \vee R(x)}_{(3) \quad (4) \quad C_2}, \underbrace{\neg Q(y) \vee R(y)}_{(5) \quad (6) \quad C_3}, \underbrace{\neg R(x) \vee W(x)}_{(7) \quad (8) \quad C_4}, \underbrace{\neg W(f(z))}_{(9) \quad C_5} \right\}$$

$C_6 = \text{Res}_{\text{lock}}(C_1, C_2) = \underbrace{Q(x)}_{(2)} \vee \underbrace{R(x)}_{(4)}$

$C_7 = \text{Res}_{\theta_1}^{\text{lock}}(C_3, C_6) = \underbrace{R(x)}_{(4)}$

$C_8 = \text{Res}_{\text{lock}}(C_4, C_8) = \underbrace{W(x)}_{(8)}$

$C_9 = \text{Res}_{\theta_2}^{\text{lock}}(C_5, C_8) = \square$

Conclusion: $S_4 \xrightarrow[\text{Res.}]{\text{lock, Pr.}} \square$, so S_4 inconsistent.

Theor. result:

⑤ inconsistent ($\Rightarrow$)

$S \vdash \frac{\text{Pr, lock}}{\text{Res.}} \square$

$\theta_1 = \text{mgu}(Q(x), Q(y))$
 $= [y \leftarrow x]$

$\theta_2 = \text{mgu}(W(f(z)), W(x))$
 $= [x \leftarrow f(z)]$